

DAEN/ISEN 427 Homework 2

Due date: Tue Sep 16, 2025 (before class)

Rubric: Full credit may be given for any reasonable Bayesian analysis.

1. From the following expressions, which one corresponds to the sentence: *The probability of being sunny given that it is June 14th of 1976?*

- (a) $p(\text{sunny})$
- (b) $p(\text{sunny} \mid \text{June})$
- (c) $p(\text{sunny} \mid \text{June 14th of 1976})$
- (d) $p(\text{June 14th of 1976} \mid \text{sunny})$
- (e) $p(\text{sunny, June 14th of 1976})/p(\text{June 14th of 1976})$

Answer: The correct answers are **c** and **e**. (the last one follows from the definition of conditional probability)

2. Show that the probability of choosing a human at random and picking the Pope is not the same as the probability of the Pope being human. In the animated series Futurama, the (Space) Pope is a reptile. How does this change your previous calculations?

Answer: Let's assume there are $7.8 \cdot 10^8$ humans and there is only one Pope (Pope Leo XIV at the time of this writing). If a human is picked at random from the entire human population, the chances of that human being the pope are 1 in $7.8 \cdot 10^8$.

$$p(\text{Pope} \mid \text{human}) = \frac{1}{7.8 \cdot 10^8}$$

Because Pope are selected among humans. We can assert that given someone is the Pope, there is 100% chance he is a human. So, $p(\text{human} \mid \text{Pope}) = 1$.

Regarding the animated series Futurama and the space Pope is a reptile. We then have that $p(\text{Pope} \mid \text{human}) = 0$, and $p(\text{human} \mid \text{Pope}) = 0$.



3. Suppose the prior distribution and the posterior distribution are as in Figure 1:

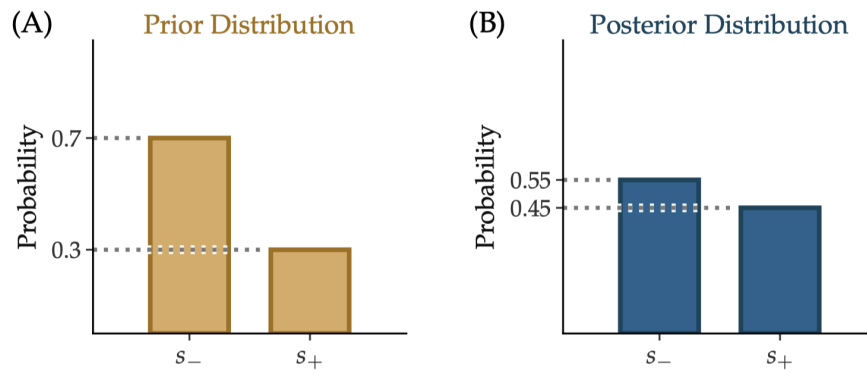


Figure 1: Prior and Posterior for Discrimination Task

(a) Calculate the likelihood ratio.

Answer: The ratio of posterior probabilities can be expressed in terms of prior ratio and likelihood ratio

$$\begin{aligned} \frac{P(s = s_+ | x)}{P(s = s_- | x)} &= \frac{P(x | s = s_+) P(s = s_+)}{P(x | s = s_-) P(s = s_-)} \\ \frac{0.45}{0.55} &= \frac{P(x | s = s_+)}{P(x | s = s_-)} \times \frac{0.3}{0.7} \end{aligned}$$

Thus we have the likelihood ratio is $\frac{21}{11}$.

(b) Does the sensory evidence alone (without the prior) indicate that the stimulus was s_+ or s_- ?

Answer: As

$$\frac{P(x | s = s_+)}{P(x | s = s_-)} = \frac{21}{11} > 1$$

the likelihood alone indicates s_+ .

(c) Do the prior and the likelihood favor the same alternative?

Answer: No, the likelihood favors s_+ while the prior favors s_- .

4. Assume the likelihood function and a posterior distribution are Gaussian (see figure 2). Assume that the prior is also Gaussian. Which of the following statements is true? Explain.

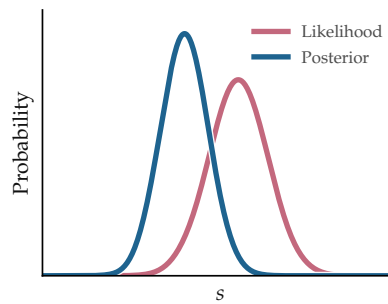


Figure 2: Gaussian distributions

- (a) The prior is centered to the left of the likelihood function and is narrower.
- (b) The prior is centered to the left of the likelihood function and is wider than it.
- (c) The prior is centered to the right of the likelihood function and is narrower than it.
- (d) The prior is centered to the right of the likelihood function and is wider than it.

Answer: Recall the mean and variance of the posterior distribution are:

$$\mu_{post} = \frac{\frac{1}{\sigma^2}}{\frac{1}{\sigma^2} + \frac{1}{\sigma_s^2}} x + \frac{\frac{1}{\sigma_s^2}}{\frac{1}{\sigma^2} + \frac{1}{\sigma_s^2}} \mu_s \quad \sigma_{post}^2 = \frac{1}{\frac{1}{\sigma^2} + \frac{1}{\sigma_s^2}}$$

Where x is the observation (mean of the likelihood distribution).

Since $\mu_{post} < x$ this implies

$$\mu_{post} = \frac{\frac{1}{\sigma^2}}{\frac{1}{\sigma^2} + \frac{1}{\sigma_s^2}} x + \frac{\frac{1}{\sigma_s^2}}{\frac{1}{\sigma^2} + \frac{1}{\sigma_s^2}} \mu_s < x$$

which is equivalent to:

$$\frac{\frac{1}{\sigma_s^2}}{\frac{1}{\sigma^2} + \frac{1}{\sigma_s^2}} \mu_s < \frac{\frac{1}{\sigma_s^2}}{\frac{1}{\sigma^2} + \frac{1}{\sigma_s^2}} x$$

that is $\mu_s < x$.

Hence only (a) and (b) can be true. Since $\sigma_{post} < \sigma$ and

$$\frac{1}{\sigma_{post}^2} = \frac{1}{\sigma^2} + \frac{1}{\sigma_s^2}$$

it follows that

$$\frac{1}{\sigma_s^2} = \frac{1}{\sigma_{post}^2} - \frac{1}{\sigma^2} < \frac{1}{\sigma_{post}^2}$$

which implies $\sigma_{post} < \sigma_s$.

However it can be that true that:

- $\sigma < \sigma_s$ (e.g. $\sigma^2 = 2, \sigma_s^2 = 4$ for $\sigma_{post}^2 = \frac{4}{3}$) or
- $\sigma > \sigma_s$ (e.g. $\sigma^2 = 2, \sigma_s^2 = 1$ or $\sigma_{post}^2 = \frac{2}{3}$)

so both (a) and (b) can be true.